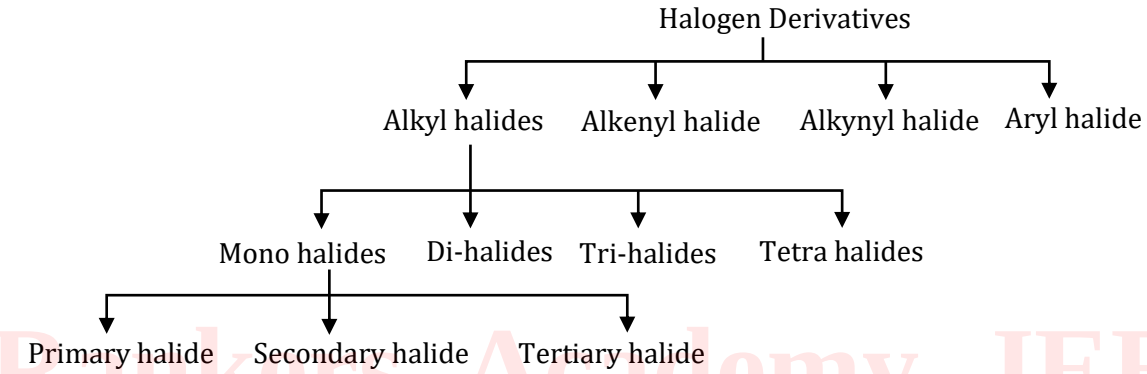
02 Alkyl Halides or Haloalkanes

Introduction:

Compounds derived from hydrocarbons by replacement of one or more H-atoms by corresponding number of halogen atoms are known as halogen derivatives.

Classification:

On the basis of nature of hydrocarbon from which they are obtained, halogen derivatives can be classified as :



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(i) Alkyl halides : Halogen derivative of alkanes.

(ii) Alkenyl halides : Halogen derivative of alkenes.

(iii)Alkynyl halides : Halogen derivative of alkynes

(iv) Aryl halides : Halogen derivative of arenes (aromatic)

Alkyl halides can be further classified on the basis of number of halogen atoms introduced in alkane molecule.

(1) Mono halides : They are obtained by replacement of one hydrogen atom by one halogen atom in alkane.

General formula $C_nH_{2n+1}X$

Example : CH_3Cl Methyl chloride (Chloro methane)

CH_3CH_2Br Ethyl bromide (Bromo ethane)

(2) Dihalides : They are obtained by replacement of two hydrogen atom by two halogen atom in alkane molecule. The two halogen atoms may be on same carbon atom, known as geminal dihalides, if two halogen atoms are at adjacent carbon atoms they are known as vicinal dihalides.

General formula $C_nH_{2n}X_2$

Example : CH_2X_2 Methylene dihalide

$\begin{matrix} CH_2X \\ | \\ CH_2X \end{matrix}$ Ethylene dihalide or Vicinal dihalide

$\begin{matrix} CH_3 \\ | \\ CHX_2 \end{matrix}$ Ethylidene dihalide or geminal dihalide

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(3) Trihalides : They are obtained by replacement of three hydrogen atom by three halogen atom in alkane.

General formula - $C_nH_{2n-1}X_3$.

Example : CHX_3 Trihalo methane or haloform

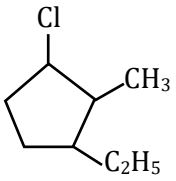
(4) Tetra halide and Perhalo compounds : They are obtained by replacement of four hydrogen atom by four halogen atom in alkane.

General formula $C_nH_{2n-2}X_4$ (tetra halide).

Example : $CH_4 \longrightarrow CX_4$ (Per halo methane)

$C_2H_6 \longrightarrow C_2H_6$ (Per halo ethane)

Illustration 1



has the IUPAC name :

(A) 3–Chloro–1–ethyl–2–methyl cyclopentane

(B) 1–Chloro–3–ethyl–2–methyl cyclopentane

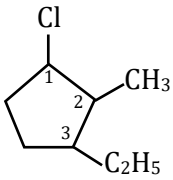
(C) 4–Chloro–1–ethyl–5–methyl cyclopentane

(D) 2–methyl–3–ethyl–1–chloro cyclopentane

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Ans. (B)

Solution:



1-chloro-3-ethyl-2-methyl cyclopentane

Illustration 2

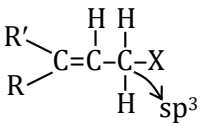
The hybridisation of carbon in C–X bond of allylic halides?

(A) sp^2 (B) sp (C) sp^3 (D) dsp^2

Ans. (C)

Solution:

Allhylic halide is

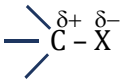


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Nucleophilic substitution reaction (S_N):

Nature of c – x bond :

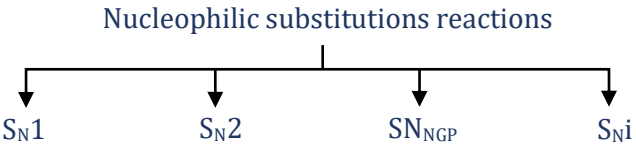
Since halogen atoms are more electronagtive than carbon, the carbon halogen bond or alkyl halide is polarised, the carbon atom bears a partial positive charge whereas the halogen atom bears a partial negative charge.



Replacement (displacement) of an atom or group by an other atom or group in molecule is known as substitution reaction. If substitution reaction is brought about by a nucleophile then it is known as nucleophilic substitution reaction. Generally substitution takes place at sp^3 carbon.

$$R\text{–}L + Nu^{\ominus} \longrightarrow R\text{—}Nu + L^{\ominus} \qquad \text{(L–leaving group)}$$

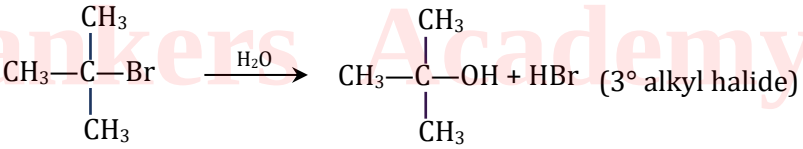
Nucleophilic substitution reactions can be classified as following :



Unimolecular nucleophilic substitution reaction (S_N1):

Mechanism : S_N1 in alkyl halides

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Nucleophilic substitution involves two step process

First step : Slow step involves ionisation to form carbocation. It is a rate determining step

$$R\text{—}L \longrightarrow R^{\oplus} + L^{\ominus} \text{(Slow)}$$

Ionisation is always assisted by the solvent since energy necessary to break the bond is largely recovered by solvation of $R^{\oplus}$ and $L^{\ominus}$

Second step : Attack of nucleophile on carbocation to result into product, it is a fast step

$$R^{\oplus} + Nu^{\ominus} \longrightarrow R\text{—}Nu \text{ (fast)}$$

1. **Characteristics of S_N1 reactions :**

   (A) It is unimolecular, two step process.

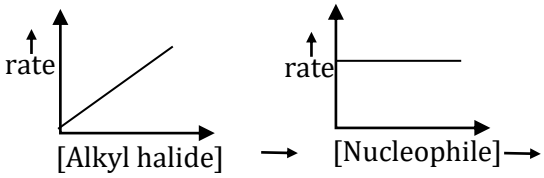
   (B) Carbocation intermediate is formed so rearrangement is possible in S_N1 reaction.

   $$\text{Rate} \propto [\text{Alkyl halide}]^1$$

   $$\text{Rate} = K\,[\text{Alkyl halide}]^1$$

   Rate of S_N1 reaction is independent of concentration and reactivity of nucleophile, i.e., It is first order reaction

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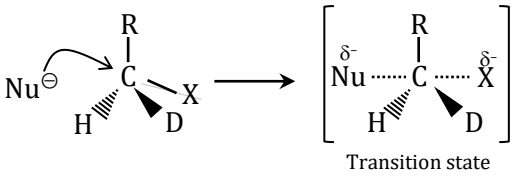
For More Details, Refer Chapter Reaction Mechanism- II

Bimolecular nucleophilic substitution reaction (S_N2):

Nucleophilic substitution in which incoming group replaces leaving group in one step only.
Reaction does not involve formation of any intermediate.

S_N2 Reaction of Alkyl halide : $R - X + Nu^{\ominus} \rightarrow R - Nu + X^{\ominus}$

Mechanism :



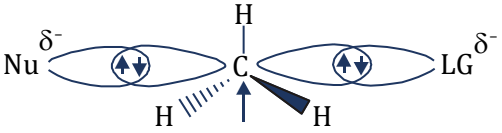
Characteristics of S_N2 reactions:

(A) Nucleophile attacks on the substrate from just opposite side to the leaving group.

(B) Hybridisation of the carbon at which substitution occurs changes from sp^3 to sp^2 in the transition state.

(C) It is bimolecular, one step concerted process
rate $\propto$ [alkyl halide] [nucleophile]
rate = k[alkyl halide] [nucleophile]

(D) Transition state has following structure



(Pentavalent carbon in sp^2 hybridised state)

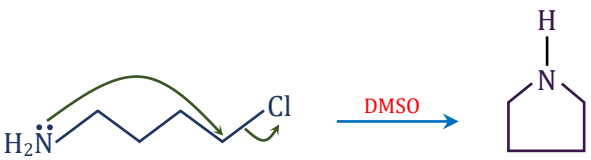
Bond with nucleophile and leaving group are relatively long and weak in transition state.

For More Details, Refer Chapter Reaction Mechanism- II

S_NNGP Nucleophilic substitution by Neighbouring Group Participation:

When substrate contains a group with an unshared pair of electron beta to the leaving group or sometimes further away then it is called neighboring group participation or S_NNGP.

Example :



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Characteristics of reaction :

S_NNGP is two step reaction-

Step I : Intramolecular S_N2 which is NGP step

Step II : Intermolecular S_N2

Compound having lone pair at 3rd, 5th or 6th atom with respect to leaving group show the NGP step (Intramolecular S_N2).

Nucleophile must be anti to leaving group to show NGP step within the same molecule.

S_NNGP is faster than S_N1 & S_N2

Preparation of alkyl halide:

1. By direct halogenation of alkanes :

$$R—H \xrightarrow{X_2/h\nu} R—X + HX$$

Reactivity of above reaction with respect to type of hydrogen to be replaced follows given order Tertiary hydrogen > Secondary hydrogen > Primary hydrogen

As far as the reactivity of halogen is concerned, F_2 is most reactive while I_2 is least reactive. Infect, reaction with I_2 is reversible and is carried out in the presence of some oxidising agents like HIO_3, HNO_3 etc. to oxidise HI into I_2.

2. By the addition of H–X on alkenes :

Alkyl chlorides, bromides and iodides can be prepared by treating an alkene with corresponding halogen acid (HCl, HBr or HI). The addition of these compounds to alkene takes place according to Markovnikov's rule. The reaction proceeds by electrophilic addition of $H^{\oplus}$ to give more stable carbocation followed by attack of $X^{\ominus}$.

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$$CH_3—CH=CH_2 \xrightarrow{HX} \underset{\text{(2° carbocation)}}{CH_3—\overset{\oplus}{C}H—CH_3} \xrightarrow{X^{\ominus}} CH_3—\underset{\displaystyle X}{\underset{|}{C}}H—CH_3$$

Anti-Markovnikov addition of HBr can be achieved, if the reaction is carried out in presence of peroxides (H_2O_2 or benzoyl peroxide). Addition of HBr to alkenes in the presence of peroxide follows free radical mechanism.

$$CH_3—CH=CH_2 \xrightarrow[\text{Peroxide}]{HBr} \underset{\text{(2° radical)}}{CH_3—\dot{C}H—CH_2—Br} \xrightarrow{HBr} CH_3—CH_2—CH_2Br + \dot{B}r$$

3. By alcohols :

(a) By the action of hydrogen halides :

Example : $R–CH_2—OH \xrightarrow[ZnCl_2]{H–X} RCH_2—X$

Mechanism :

$$R—CH_2—\ddot{O}H \underset{(H–X)}{\overset{H^{\oplus}}{\rightleftharpoons}} \underset{\displaystyle H}{\underset{|}{R—CH_2—\overset{\oplus}{O}—H}} \xrightarrow{-H_2O} R—\overset{\oplus}{C}H_2 \xrightarrow{X^{\ominus}} \underset{\text{(Product)}}{R—CH_2—X}$$

(unstable)

In this reaction intermediate carbocation is formed so rearrangement ($H^{\ominus}$ shifting or $CH_3^{\ominus}$ shifting) can take place.

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$ZnCl_2$ act as dehydrating agent and absorbs H_2O from the reaction so good yield of halide is obtained. Also it generates $H^{\oplus}$ from HCl.

$$HCl + ZnCl_2 \longrightarrow ZnCl_3^{\ominus} + H^{\oplus}$$

Reactivity order for alcohol :

Reactivity $\propto$ stability of intermediate carbocation

so reactivity order : **Tert. alc. > Sec. alc. > Pri. alc.**

Reactivity order of H—X is : **HI > HBr > HCl**

HI reacts readily with 1°, 2° and 3° alcohols.

$$R—OH + HI \longrightarrow R—I + H_2O$$

HCl and also 1° alcohol are less reactive so $ZnCl_2$ or some amount of H_2SO_4 is needed to increase the reactivity.

Example : $CH_3—CH_2—OH + HCl \xrightarrow{ZnCl_2} CH_3—CH_2—Cl$

At normal condition (without $ZnCl_2$) :

$$CH_3—CH_2—OH + HCl \longrightarrow \times \text{ (no reaction)}$$

Note : $HCl + ZnCl_2$ is called as **lucas reagent,** alcohol gives turbidity with lucas reagent.

Reactivity towards lucas reagent (difference in 1°, 2° and 3° alcohol).

	1° alcohol	2° alcohol	3° alcohol
Time to give turbidity	in 30 min.	in 5 min.	immediate

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(b) By the action of phosphorus halides :

$$R-OH+PCl_5 \longrightarrow R-Cl+POCl_3+HCl$$

$$3R-OH+PCl_3 \longrightarrow 3R-Cl+H_3PO_3$$

$$3R-OH+PBr_3 \longrightarrow 3R-Br+H_3PO_3$$

$$3R-OH+PI_3 \longrightarrow 3R-I+H_3PO_3$$

Phosphorous halides are prepared by treating red phosphorous and halogen. The advantage of using phosphorous halides is that the reaction does not involve carbocation intermediate so, it is free from rearrangement.

PBr_3 and PI_3 are less stable, thus for bromides and Iodides, $(P + Br_2)$ Or $(P + I_2)$ mixture is used.

Reaction with thionyl chloride in presence of pyridine:

Thionyl chloride ($SOCl_2$) is often the best reagent for converting an alcohol to an alkyl chloride. The by products (gaseous SO_2 and HCl) leave the reaction mixture and ensure that there can be no reverse reaction.

$$R—OH + Cl—\overset{\overset{O}{\|}}{S}—Cl \xrightarrow[\text{Pyridine}]{\text{Heat}} R—Cl + SO_2 + HCl$$

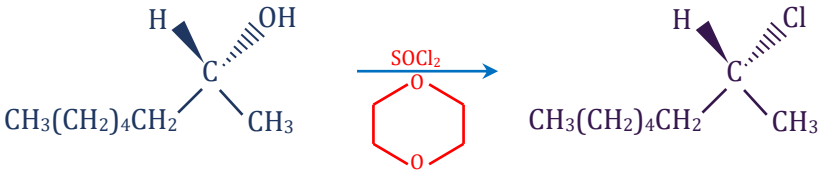
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Reaction with thionyl chloride:

$ROH + SOCl_2 \longrightarrow RCl + SO_2 + HCl$

In this mechanism an internal nucleophile attacks from the same side of leaving group, means retention of configuration. It is an S_Ni mechanism, where i means internal

Example :



Finkelstein reaction:

$$R—OH + HI \longrightarrow R—I \xrightarrow{HI} R—H + I_2$$

(reduced) [Final product is Alkane (R-H) not R-I]

This is a problems that is why iodides are best prepared through halogen exchange reaction. It is also known as Finkelstein reaction. In this reaction R—Cl and R—Br is treated with sodium iodide in acetone.

$$Ph—CH_2—Br \xrightarrow[Acetone]{NaI} \underset{(Soluble)}{Ph—CH_2—I} + NaBr \downarrow$$

$$R—Br \xrightarrow[Acetone]{NaI} R—I + NaBr \downarrow$$

- In this reaction acetone is used because sodium iodide (NaI) is soluble in acetone but NaBr and NaCl are insoluble so precipitated out. This eliminates any possibility of reverse reaction.

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- It is an S_N2 reaction therefore only 1° RX and 2°RX is used.



Swarts Reaction:

The synthesis of alkyl fluorides is best accomplished by heating an alkyl chloride/ bromide in presence of metallic fluoride such as

$$AgF, Hg_2F_2, CoF_2 \text{ or } SbF_3.$$

$$CH_3–X + AgF \rightarrow CH_3–F + AgX$$

$$X = Br, Cl, I$$

$\Rightarrow$ F^- is a good nucleophile in polar aprotic solvent hence this reaction occurs via S_N2 mechanism.

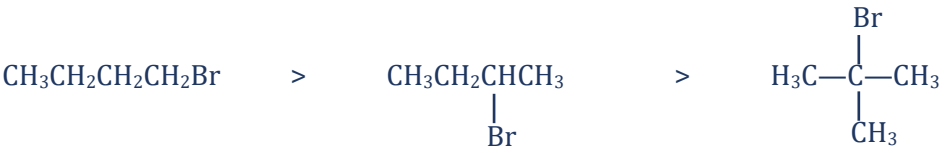
Physical Properties of alkyl halides:

(i) **Boiling points and Melting points :**

For the same alkyl group, the boiling points of alkyl halides decrease in the order: RI> RBr> RCl> RF. This is because with the increase in size and mass of halogen atom, the magnitude of van der Waal forces increases. The boiling points of isomeric haloalkanes decrease with increase in branching.

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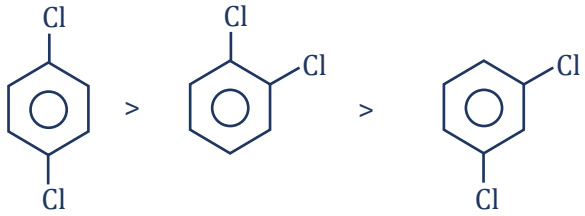
Boiling point order :



Boiling points of isomeric dihalobenzenes are nearly the same.

However, the para-isomers has high melting as compared to their ortho- and meta-isomers. It is due to symmetry of para-isomers that fits in crystal lattice better as compared to ortho- and meta-isomers.

Melting point order :



(ii) **Density :** The density increases with increase in number of carbon atoms, halogen atoms and atomic mass of the halogen atoms.

(iii) **Solubility :** The haloalkanes are slightly soluble in water. However, haloalkanes tend to dissolve in organic solvents because the new intermolecular attractions between haloalkanes and solvent molecules have much the same strength as the ones being broken in the separate haloalkane and solvent molecules.

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(iv) The lower members CH_3F, CH_3Cl, CH_3Br, C_2H_5Cl and C_2H_5F are gases at room temp.

CH_3I and members upto C_{18} are colourless sweet smelling liquids.

(v) Higher B.P. than parent alkanes.

Decreasing order of B.P. is : **R – I > R—Br > R—Cl > R—F**

among isomeric R—X decreasing order of B.P. is : **Primary > Secondary > tertiary**

(vi) R—F and R—Cl $\longrightarrow$ lighter than water

R—Br and R—I $\longrightarrow$ heavier than water

Decreasing order of density is : **R—I > R—Br > R—Cl > R—F**

(vii) R—X are polar co-valent compounds but insoluble in water because they can not form H–bonds. They dissolve in organic solvents.

(viii) R—X burns with a green flame due to interaction of X with Cu wire.(Beilstein test)

(ix) The stability order is : **R—F > R—Cl > R—Br > R—I**

R—I is least stable and darken in light due to photodecomposition.

$2R—I \longrightarrow R—R + I_2$

(x) Lower members are colourless, oily liquids with sweet smell. Higher members are solid.

(xi) The reactivity of gemdihalides is lesser than vicinal or mono halides. [Reason - in presence of one halogen atom (Strong attracting –I effect) the other halogen atom can not be so easily replaced.

(xii) These are heavier than water.

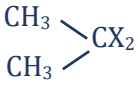
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Dihalides:

General formula $C_nH_{2n}X_2$. Two H - atom of alkanes, replaced by two halogen atoms to form dihalides.
Dihalides are classified as :

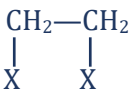
(a) Gem dihalide : The term Gem is derived from geminal means - same position.
Two similar halogen atoms are attached to same C - atom

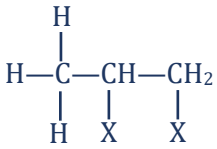
Example : CH_3CHX_2 ethylidene dihalide (1, 1 - dihalo ethane)


Isopropylidene dihalide (2, 2 - dihalo propane)

(b) Vic dihalides : Vic term from - Vicinal means adjacent C - atoms
Two halogen atoms are attached on adjacent carbon atom.

Example :

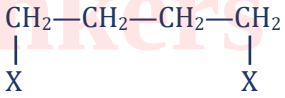

ethylene dihalide
(1,2–dihaloethane)


Propylene dihalide
(1,2–dihalopropane)

Vic and Gem dihalides are position isomers.

(c) α, δ dihalides : Halogen atoms are attached with terminal C - atom. They are separated by 3 or more C - atom . They are also known as polymethylene halides.

Example :

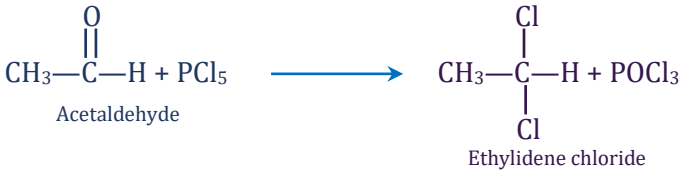

(1,4–dichloro butane)
Tetramethylene dichloride

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General method of preparation:

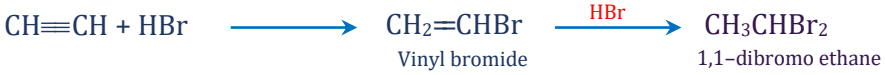
(a) Gem dihalides :

(i) By the reaction of PCl_5 on carbonyl compound.

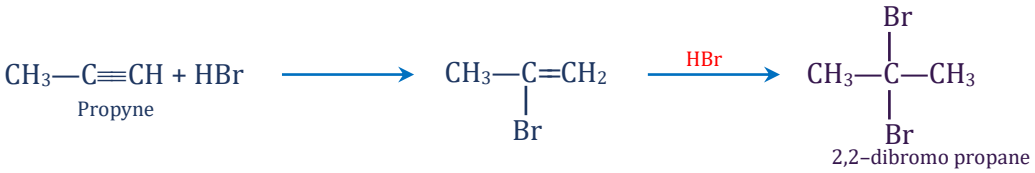

Acetaldehyde → Ethylidene chloride


Acetone → 2,2–dibromo propane

(ii) By addition of halogen acids on alkynes :


Vinyl bromide → 1,1–dibromo ethane

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(b) Vic-dihalides :

(i) By the addition of halogens to alkenes :





(ii) By the action of PCl_5 on glycols :

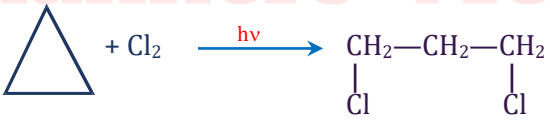


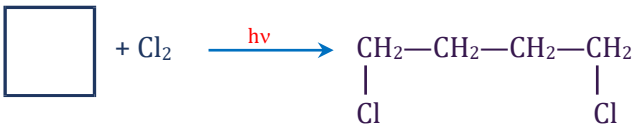
(c) α – δ dihalides :

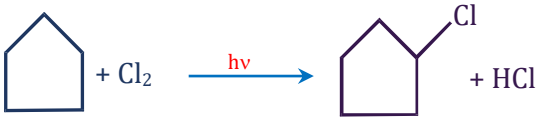
(i) $CH_2CH—CH_2Br + HBr \longrightarrow Br—CH_2—CH_2—CH_2Br$

1, 3 - dibromo propane

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(ii) 

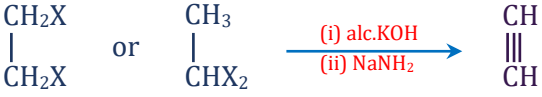
(iii) 

(iv) 

Note : Here  has less tendency to open the ring.

Chemical properties:

(i) Action of KOH(alc.) : (Dehydrohalogenation)



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(ii) Action of KOH(aq.) : (Hydrolysis) It is a distinction test for gem and vic dihalides.

(a) $CH_2Cl-CH_2Cl + 2KOH(aq.) \longrightarrow CH_2OH-CH_2OH + 2KCl$

Vic– dihalide — Glycol 1,2–ethane diol.

(b) $CH_3CHXCH_2X \xrightarrow{KOH(aq.)} CH_3-CH(OH)-CH_2OH$

Vic– dihalide — Propane 1,2–diol.

Gem. $CH_3CHX_2 \xrightarrow{KOH(aq.)} CH_3CHO$ and $CH_3CX_2CH_3 \xrightarrow{KOH(aq.)} CH_3COCH_3$ (ketone)

(iii) Reaction with KCN : Gem and Vic dihalide gives different products

Vic. :

$CH_2Cl-CH_2Cl + 2KCN \xrightarrow{-2KCl} CH_2CN-CH_2CN \xrightarrow{H_2O/H^+} CH_2COOH-CH_2COOH \xrightarrow[-H_2O]{\Delta}$ Succinic anhydride

Succinic acid

Succinic anhydride

Gem. :

$CH_3-CH(Cl)_2 \xrightarrow[-2KCl]{2KCN} CH_3-CH(CN)_2 \xrightarrow{H_2O/H^{\oplus}} CH_3-CH(COOH)_2 \xrightarrow[-CO_2]{\Delta} CH_3-CH_2COOH$ Propionic acid

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(a) – CN group on acid hydrolysis gives - COOH

(b) Two – COOH group on one C – atom on heating always loose CO_2 to form monocarboxylic acid.

(c) Two – COOH group on vic. C – atom on heating loose H_2O to form anhydride.

(iv) Dehalogenation :

Vic. $CH_2Br-CH_2Br + Zn \xrightarrow[Heat]{CH_3OH} CH_2=CH_2 + ZnBr_2$

Same carbon product

Gem. $CH_3-CH_2CHBr_2 + Zn \longrightarrow CH_3-CH_2-CHCH-CH_2-CH_3 + ZnBr_2$

(Twice Carbon Product)

α – γ. $CH_2(CH_2Br)_2 + Zn \xrightarrow[\Delta]{CH_3OH} H_2C(CH_2)_2 + ZnBr_2$

α, γ–dihalide — cyclopropane

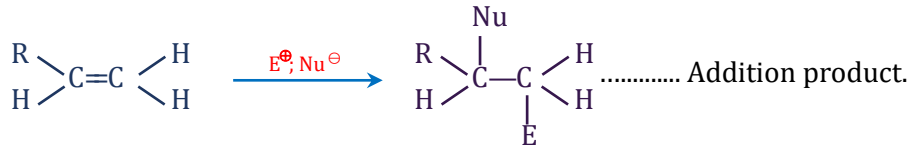
ELECTROPHILIC ADDITION REACTION:

(I) Electrophilic addition reaction of Alkenes:

Due to presence of weak π electrons in alkene and alkyne, it will go for electrophilic addition reaction. In electrophilic addition reactions one weak π-bond (251 KJ mol^{-1}) is broken and two strong σ bond ($2 \times 347 = 694$ KJ mol^{-1}) are formed. The overall reaction is accompanied by a release of about 694–251 = 443KJmol^{-1} of energy.

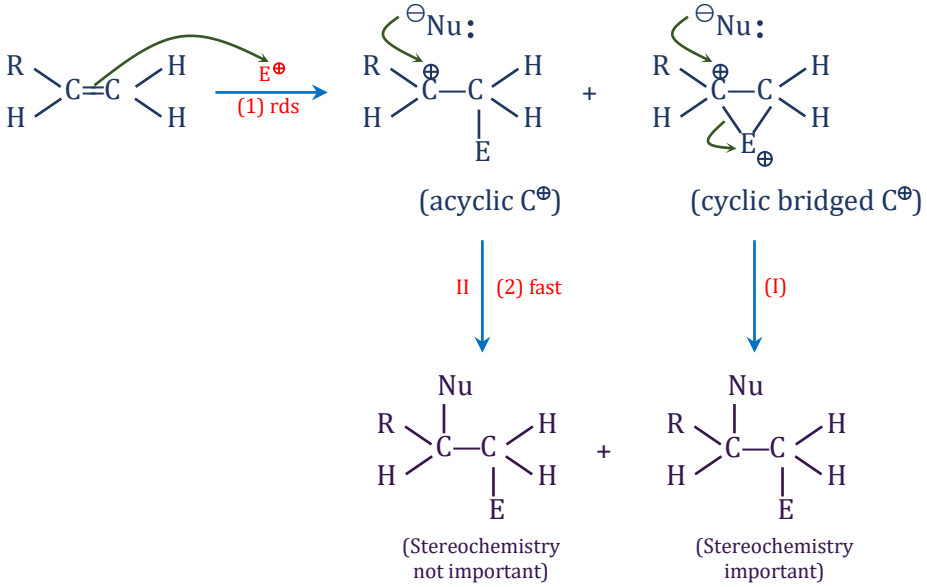
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General Reaction of electrophilic addition :



..... Addition product.

The overall reaction mechanism can be visualised as followed :



Anti Addition :

Remarks :

Alkenes, Alkynes and Alkadienes are electron rich species. So they function as $Nu^{\ominus}$ species (due to loosely bound π-electrons)

These compounds mainly give electrophilic addition reactions. Due to nucleophilic nature of alkenes/alkynes having affinity for $E^{\oplus}$.

The reaction is initiated by an attack of $E^{\oplus}$.

Rate of reaction $\propto$ stability of carbocation

Reactivity of an Alkene :

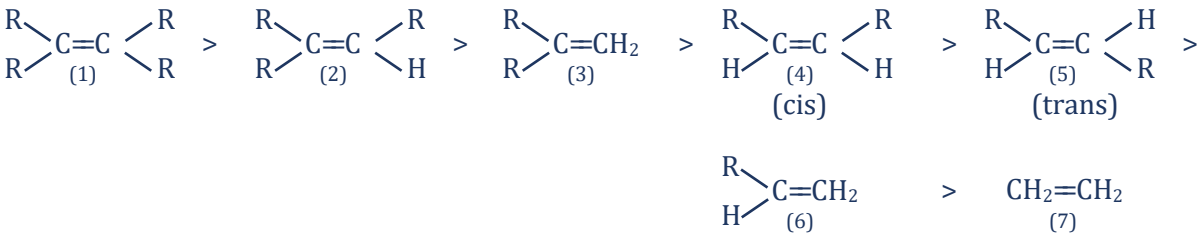
(1) Presence of e- releasing groups (+m, +I) at C = C increases nucleophilicity and reactivity.

(2) Presence of ERG stabilises the intermediate carbocation.

(3) More stable $C^{\oplus}$, more is reactivity of alkene.

Examples of Reactivity Orders :

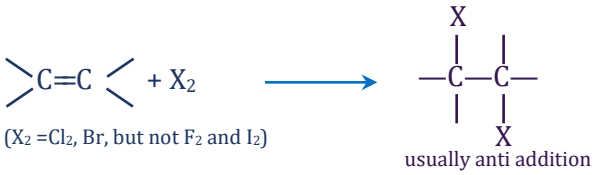
Ex :



Example : ERG – $CH=CH_2$ > $CH_2=CH_2$ > EWG – $CH=CH_2$

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(i) Addition of Halogen (Halogenation) :

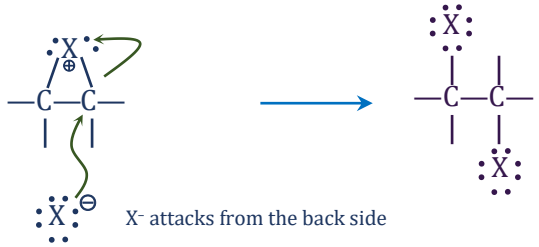


Mechanism :

Step 1 :



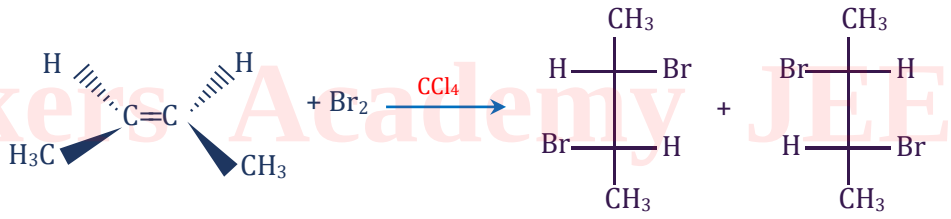
Step 2 :



Remarks : Anti addition; No rearrangement; Bridged Carbocation; Markovnikov's additions

The addition of bromine to cyclopentene is a stereospecific **anti addition.**

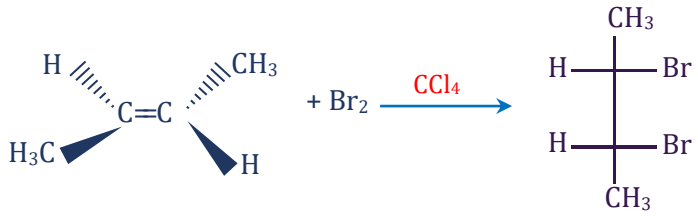
Example :



cis–2–butene

(±)–2,3–dibromobutane
(Racemic mixture)

Example :



Trans–2–butene

Meso–2,3–dibromobutane

(ii) Addition of hox :

Reagents : (i) $X_2 + H_2O \rightarrow HOX + HX$ or (ii) HOX , $H^{\oplus}$

Electrophile : $X^{\oplus}$, Halonium ion ;

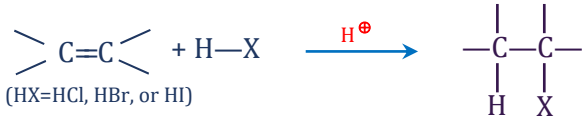


Remarks : Anti addition ; No rearrangement ; Bridged Carbocation ; Markovnikoff's additions

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(iii) Addition of hydrogen halides (+ hx) :

Reagents : HX; **Electrophile :** $H^{\oplus}$



(HX=HCl, HBr, or HI)

Remarks : Markovnikov's addition ; Intermediate carbocation ; Rearrangement may take place

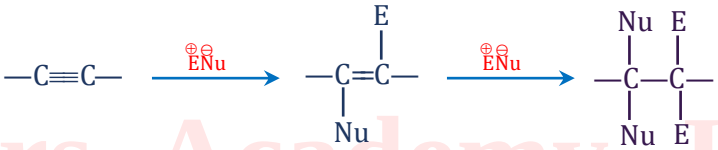
(II) Electrophilic addition reaction of Alkynes:

Many of the reactions of alkynes are similar to the corresponding reactions of alkenes. Like the pi bond of an alkene, the pi bonds of an alkyne are electron-rich, and they readily undergo addition reaction. The bond energy of the alkyne triple bond is about 226 kJ (54 kcal) more than the bond energy of an alkene double bond.

Since sigma bonds are generally stronger than pi bonds, the reaction is usually exothermic. Alkynes have two pi bonds, so upto two molecules can add across the triple bond.

We must consider the possibility of a double addition whenever a reagent adds across the triple bond of an alkyne. Some conditions may allow the reaction to stop after a single addition, while other conditions give double addition.

General Reaction :

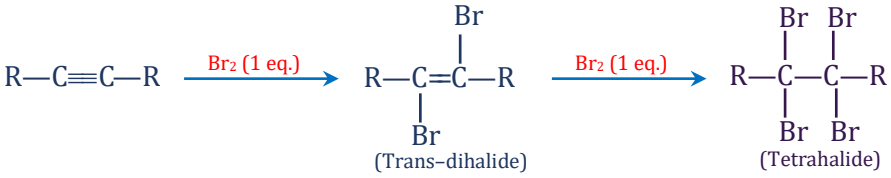


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Markovnikov's rule of addition :

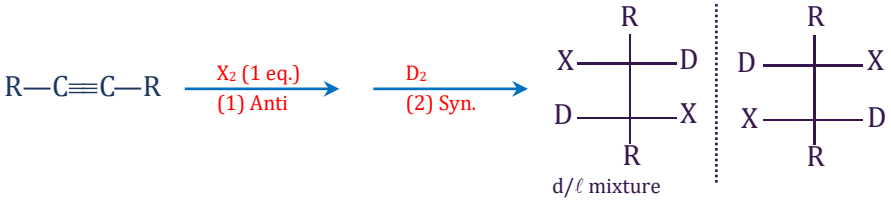
In an electrophilic addition reaction to alkenes and alkynes the electrophile attacks in such a way that a more stable carbocation intermediate is formed on which nucleophile attacks in the next step.

(i) Additon of halogen (halogenation) :

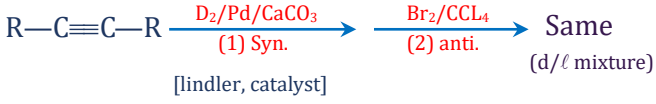


(Trans–dihalide) (Tetrahalide)

Remark : - Reaction is Anti in both step



d/ℓ mixture

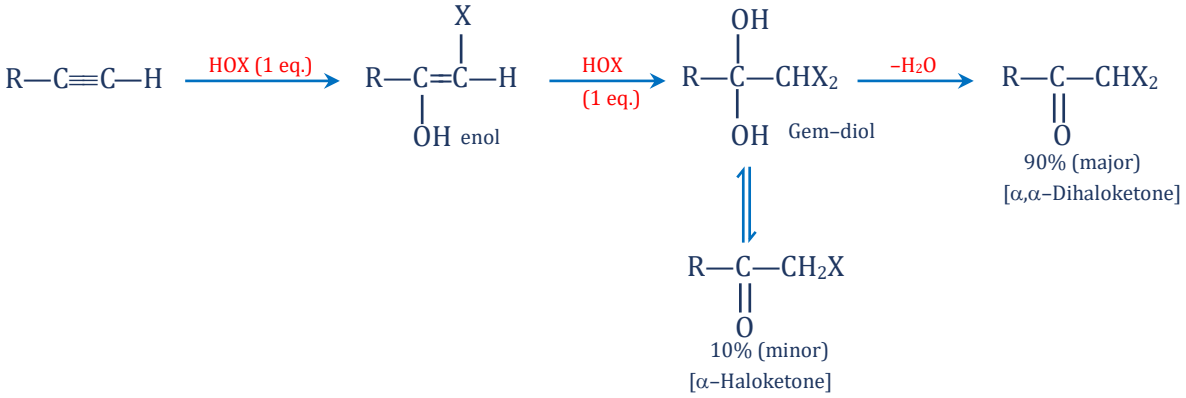


[lindler, catalyst]

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(ii) Addition of HOX :

Alkynes + HOX ⟶ α, α-dihaloketone + α-haloketone
(90% major) (10% minor)

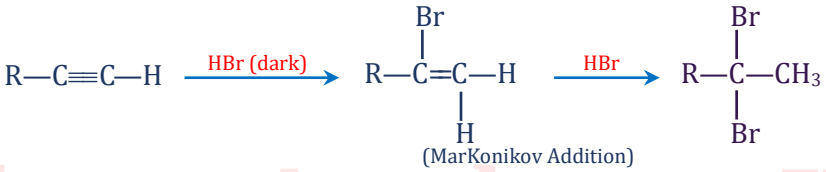


Remarks :

Two molecules of HOX can be added, the end product is α, α-Dihaloketone.

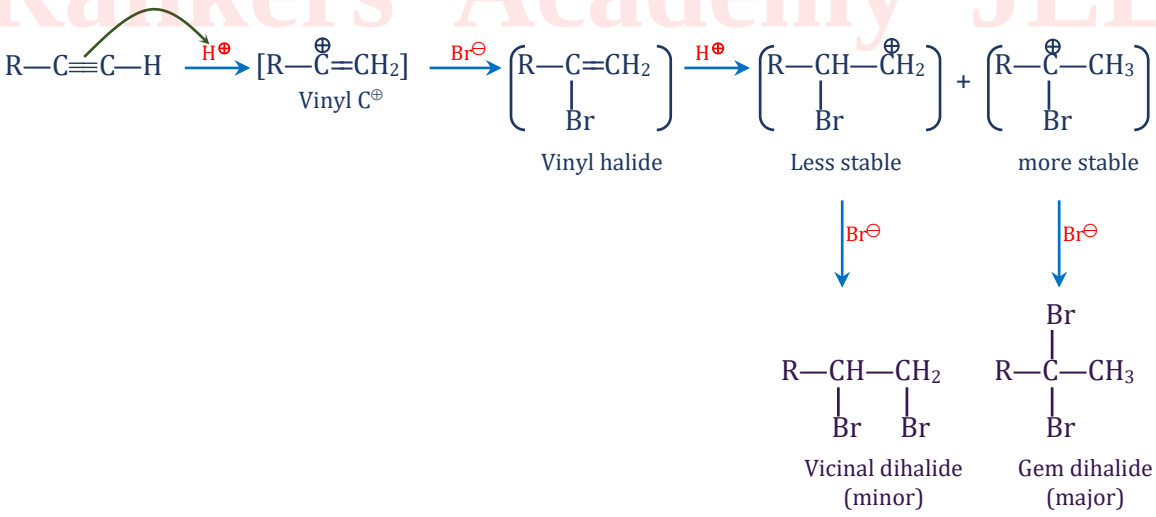
The intermediate product is an enol which gives a minor product α-haloketone.

(iii) Addition of hydrogen halides (+ HX) :



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Mechanism :



Remarks : (1) Markovnikov's Addition in both steps.
(2) If two moles of HX are added the final product is Gemdihalide.
(3) Electrophilic addition to terminal alkyne is regioselective.

1. Wurtz Reaction :

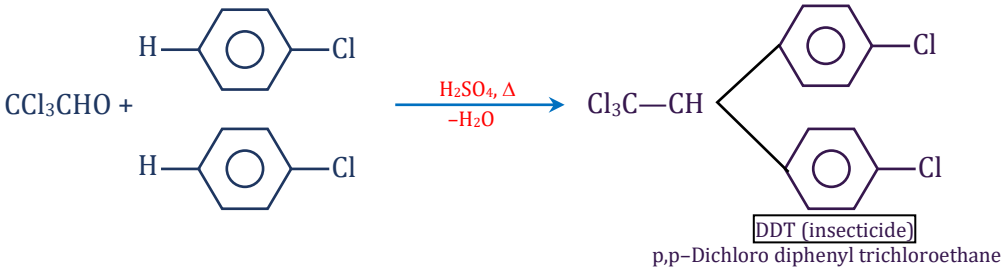
When a mixture of different alkyl halides, (R_1 - X) and (R_2 - X) is used a mixture of alkane is formed -

$$R_1—X + 2Na + X—R_2 \xrightarrow{\text{Ether}} R_1—R_2 + R_1—R_1 + R_2—R_2 + NaX$$

If 'Zn' is used in place of 'Na' the reaction is called **Frankland's Reaction**.

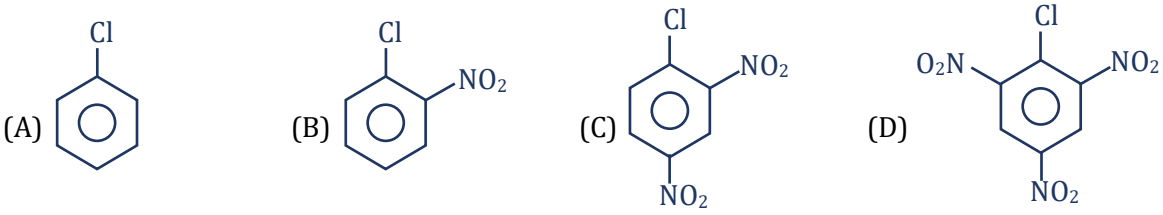
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◆ **With Chloral :**



SOLVED EXAMPLES

1. Which of the following undergoes Hydrolysis most easily :



Ans. **(D)**

Sol. If there is more m-directing group then there will be more nucleophilic substitution reaction.

2. The most reactive towards S_N1 is :

(A) $PhCH_2Cl$ (B) Ph–Cl

(C) $CH_3CHCl(CH_3)$ (D) p–NO_2—Ph—CH_2—Cl

Ans. **(A)**

Sol. The intermediate carbocation is formed.

$C_6H_5—CH_2Cl \longrightarrow$ is maximum stable due to resonance.

3. Which of the following is used as insecticide :

(A) D.D.T. (B) Chloritone

(C) Chloropicrin (D) (A) and (C) both

Ans. **(D)**

Dichloromethane :

(i) Dichloromethane is a geminal organic compound and is also called methylene chloride or methylene dichloride.

(ii) It can be obtained naturally from oceanic sources, macroalgae, volcanoes, and wetlands.

(iii) The majority of methylene dichloride in the environment is due to industrial emissions.

(iv) The chemical formula of dichloromethane is CH_2Cl_2.

Uses of Dichloromethane :

(i) Dichloromethane is used as a solvent in food technology.

(ii) It is used in aerosol formulations.

(iii) It is an ethane foam blowing agent.

(iv) It is used as solvent in the manufacturing of pharmaceutical products.

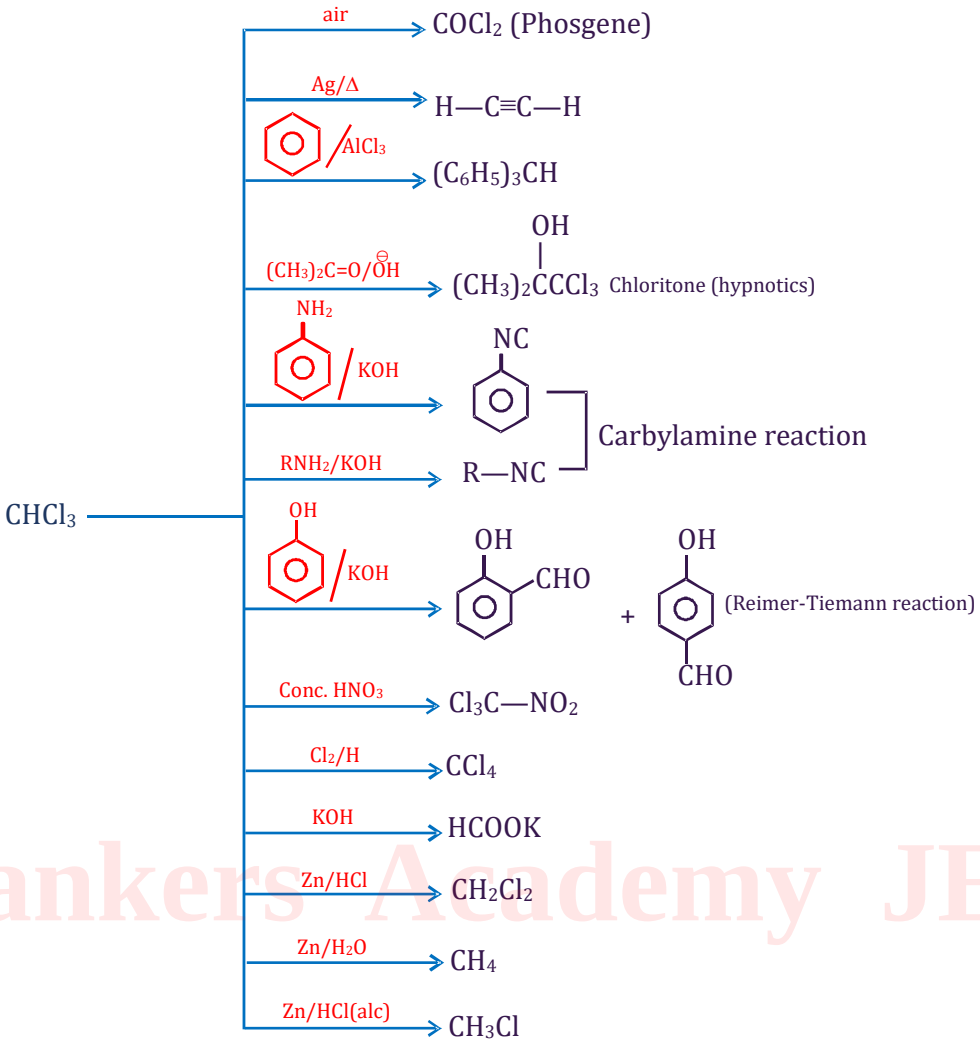
(v) CH_2Cl_2 is used as a degreasing agent.

(vi) It is used in the manufacturing of electronics.

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Reactions of chloroform:

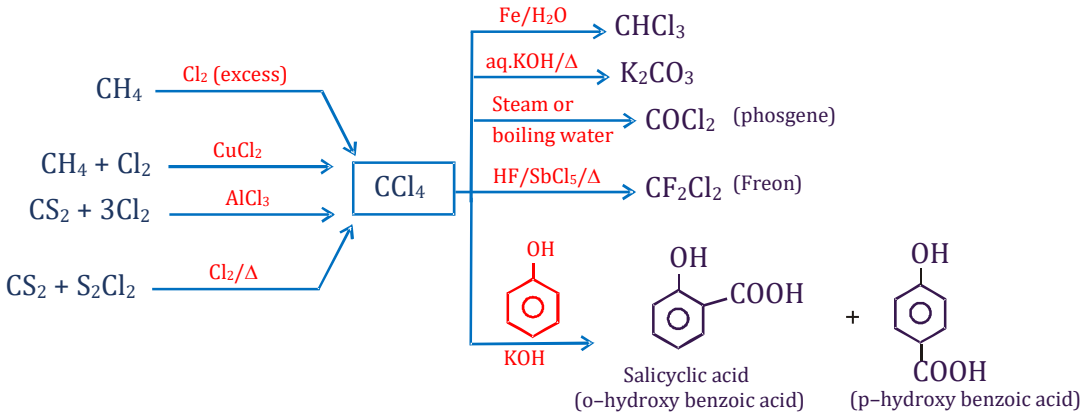


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- Purity of chloroform (presence of phosgene) can be tested before use as anaesthetic by treating with aqueous solution of $AgNO_3$ because the presence of $COCl_2$ may cause cardiac failure.
- Chloroform is stored in dark colour bottle containing small amount of ethyl alcohol. (It converts phosgene into diethylcarbonate).

Carbon tetrachloride (Pyrene):

Preparations & reactions :



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- CCl_4 is stable to heat and its vapours do not catch fire thus it is used as fire extinguisher.
- CCl_4 used as medicine for elimination of hook worms.

Freons (Polychlorofluoro Alkane):

(i) $CCl_4 + HF \xrightarrow{SbCl_4F} CCl_3F + HCl$ (Freon–11)

(ii) $CCl_3F + HF \longrightarrow CCl_2F_2 + HCl$ (Freon–12)

(iii) $C_2Cl_6 + 2HF \longrightarrow C_2Cl_4F_2 + 2HCl$ (Freon–112)

Nomenclature of Freons :

Freon is expressed as
Freon - cba
Where
a – [Number of F-atoms]
b – [1 + Number of H–atoms]
c – [Number of C-atoms –1]

Example : $C_2Cl_4F_2$ is expressed as Freon-112
a = 2 ; b = 1 + 0 ; c = 2 – 1
cab = 112

Freon is odourless, non-corrosive, non-toxic gas. It can easily be liquefied hence widely used as refrigerant.

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Iodoform Test:

CH_3CH_2OH or CH_3COCH_3 $\xrightarrow[\text{or } I_2 + Na_2CO_3 + H_2O]{I_2 + NaOH \text{ or } NaOI \text{ or } OI^{\ominus}}$ CHI_3 Yellow crystals of iodoform

Statement : CHI_3 give yellow ppt. of AgI with $AgNO_3$ but $CHCl_3$ does not give AgCl ppt.
Reason : CHI_3 is thermally less stable than $CHCl_3$.

Iodoform test can be used to distinguish the following pairs of compounds.

(i) CH_3CH_2OH and CH_3OH
(ii) CH_3CHO and CH_3CH_2CHO
(iii) 2–Pentanol and 3–Pentanol
(iv) Acetophenone and benzophenone
(v) 2–Propanol and 1–Propanol

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